Hashtag Community Network Analysis

A Research Study on Instagram Hashtag Co-occurrence Structure

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Abstract

This study models social media engagement through hashtag co-occurrence networks. We construct a weighted undirected graph where nodes represent hashtags and edges represent joint usage in high-engagement posts. Louvain community detection reveals thematic clusters, while centrality metrics are compared against average like counts to assess whether structural prominence aligns with audience response. The pipeline is fully reproducible and supports both scraped Instagram metadata and synthetic demonstration datasets.

1. Introduction

Hashtags function as lightweight semantic anchors that connect posts to topics, audiences, and discovery channels. Marketers and creators often combine multiple hashtags to maximize reach, which produces stable co-usage patterns over time. Understanding these patterns as a network can reveal community structure, bridge tags that connect themes, and engagement signals tied to network position.

This repository implements two complementary tracks: a Flask image captioning application for content-aware hashtag suggestions, and a graph analytics research pipeline for community detection and centrality analysis.

2. Methodology

Data preparation begins with post-level hashtag combinations. Each post contributes unordered hashtag pairs above a configurable like threshold. Pair frequencies are aggregated into edge weights. We build a weighted NetworkX graph, apply Louvain community detection, and compute weighted degree, betweenness, and eigenvector centrality. Pearson and Spearman correlations evaluate the relationship between centrality and average likes.

Metric	Value
Data source	synthetic
Nodes	49
Edges	1076
Graph density	0.9150
Average degree	43.92
Connected components	1
Communities detected	6
Modularity	0.2885

3. Results

The following figures summarize network topology, community composition, and engagement correlations produced by the analysis pipeline.

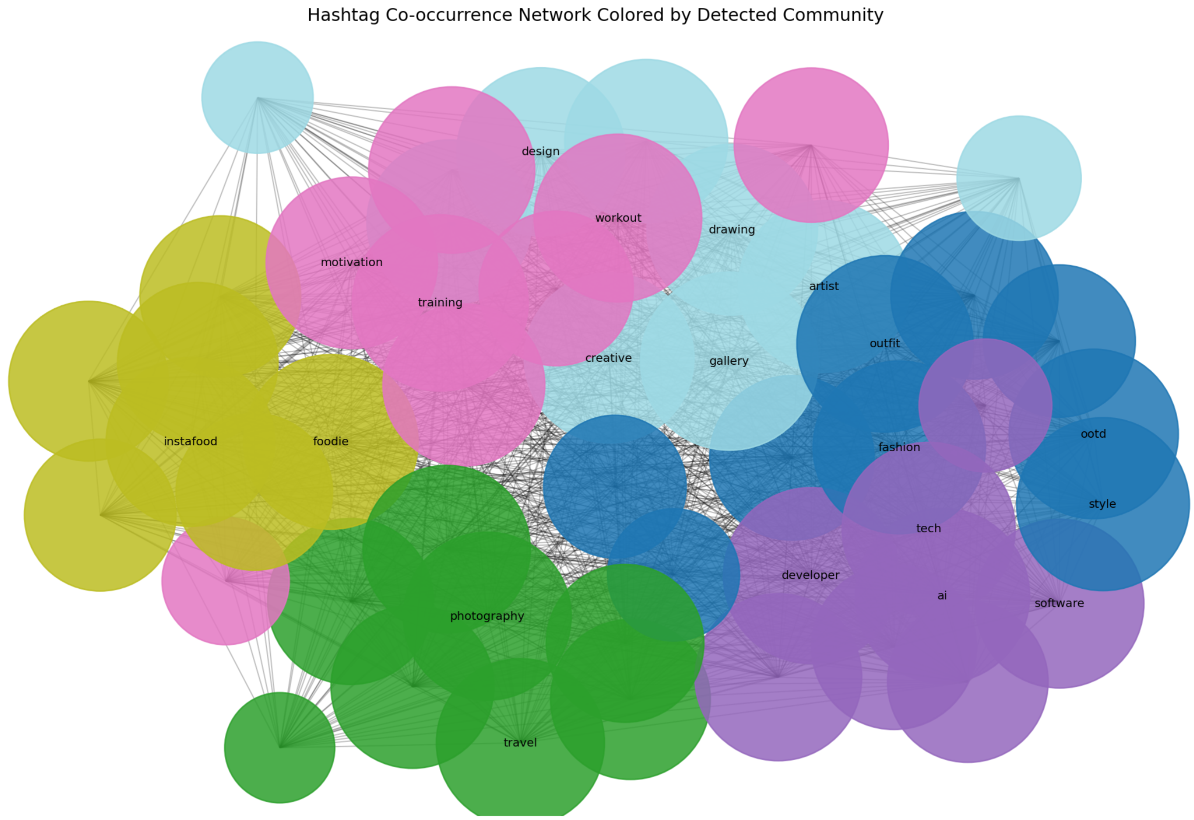

Figure 1. Hashtag co-occurrence network colored by Louvain community.

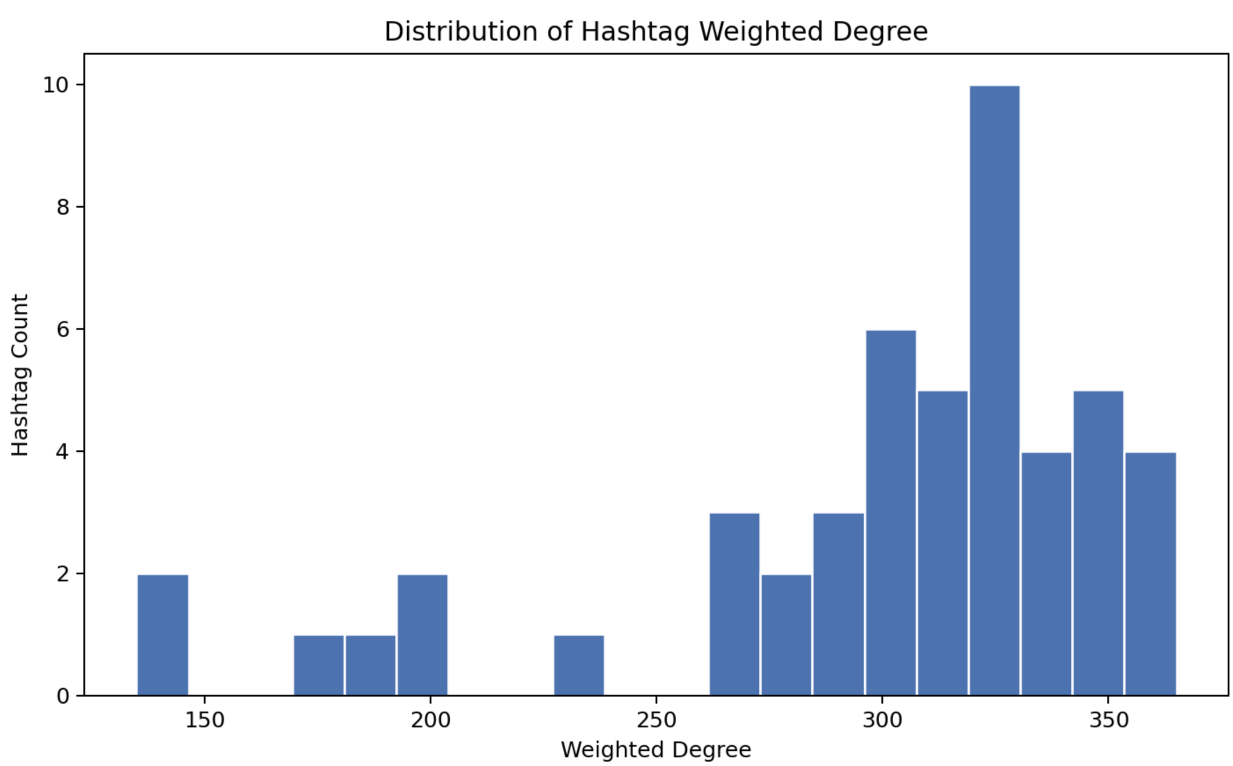

Figure 2. Weighted degree distribution across hashtags.

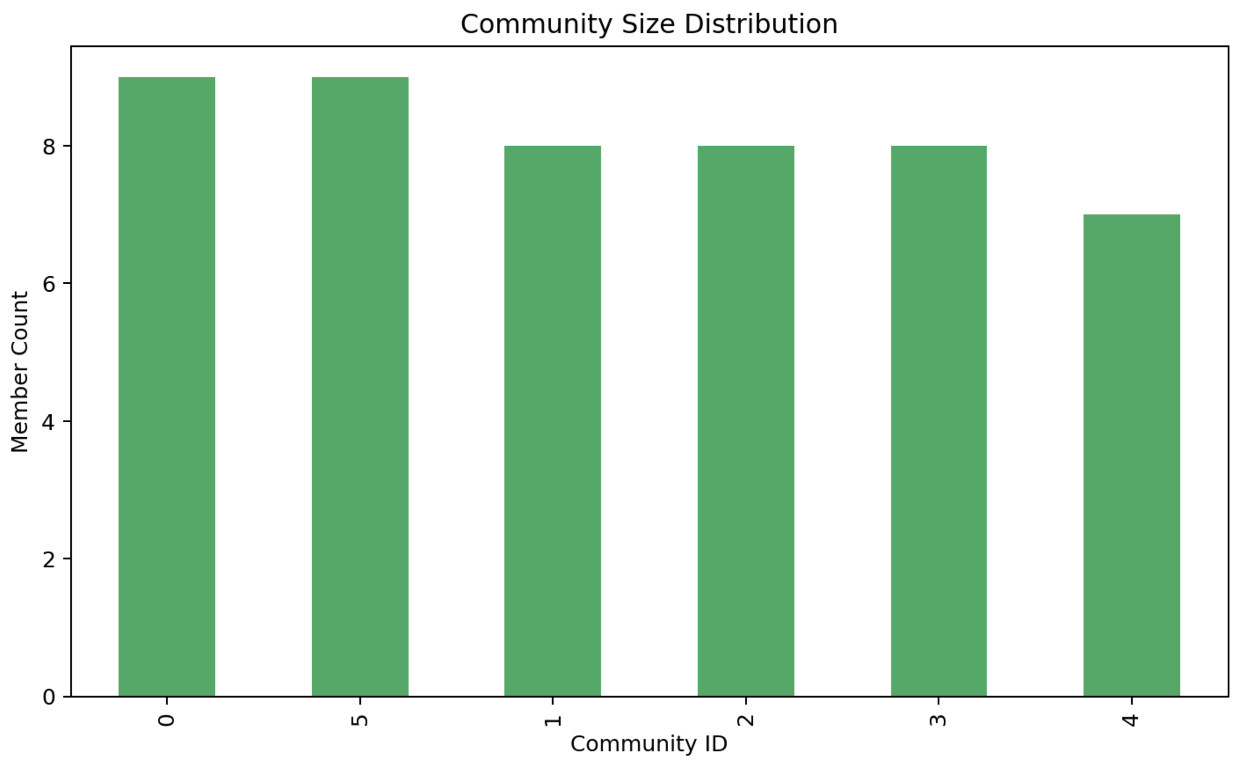

Figure 3. Community size distribution.

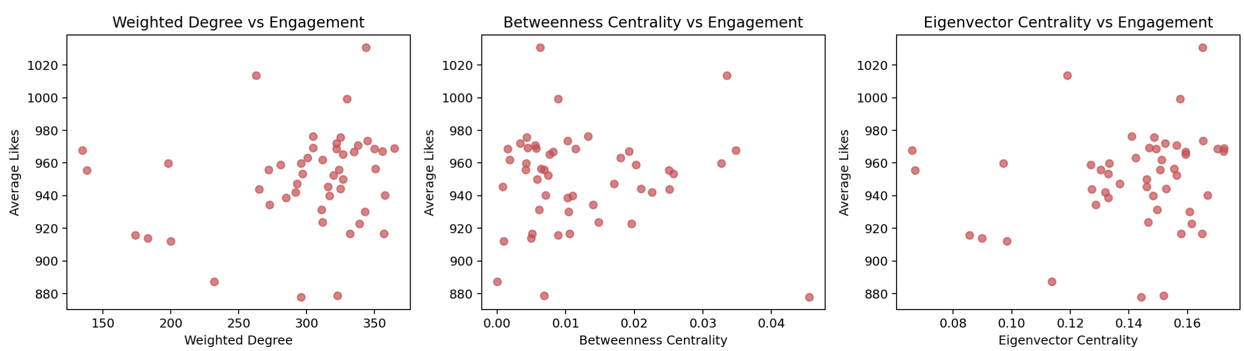

Figure 4. Centrality metrics versus average likes.

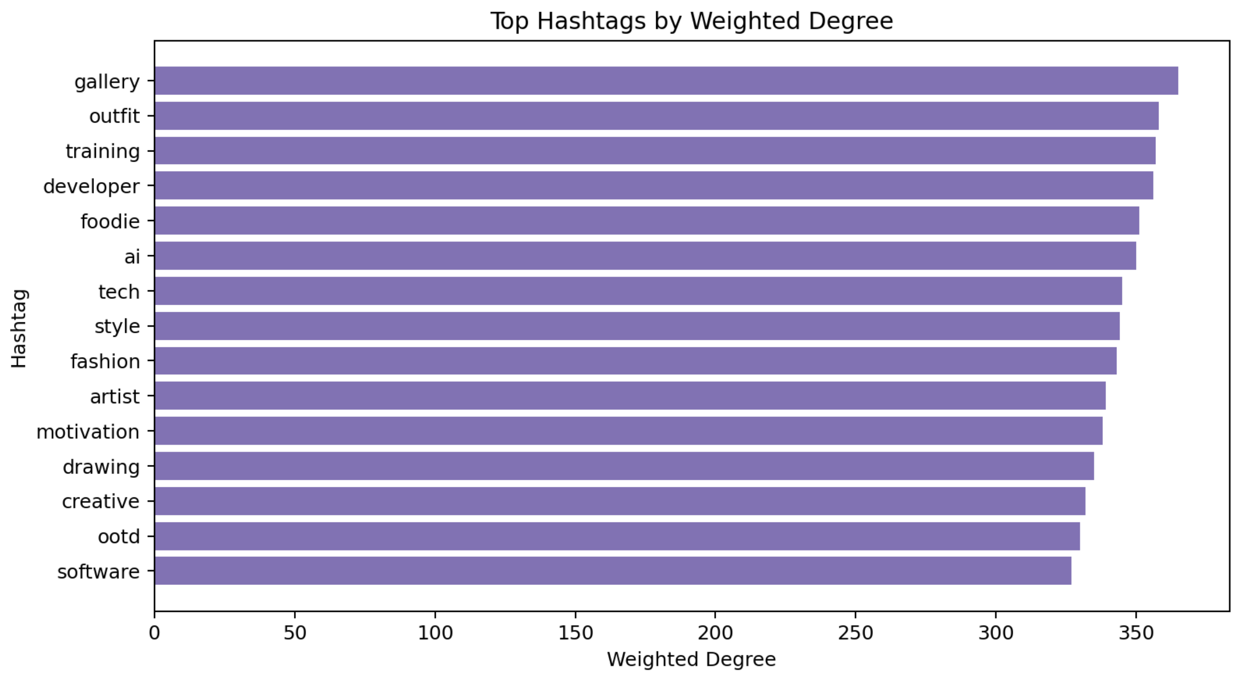

Figure 5. Top hashtags ranked by weighted degree.

4. Discussion

Centrality-engagement correlations indicate whether structurally central hashtags coincide with higher average likes. weighted_degree shows Pearson r=0.197 (p=0.175) and Spearman rho=0.252. betweenness shows Pearson r=-0.014 (p=0.923) and Spearman rho=-0.034. eigenvector shows Pearson r=0.180 (p=0.215) and Spearman rho=0.247.

Community partitions typically align with topical clusters such as travel, food, fitness, or technology when synthetic or domain-specific seed tags are present. Bridge hashtags with high betweenness can connect otherwise separate communities and are candidates for cross-topic campaigns.

5. Conclusion

Graph-based hashtag analysis provides actionable structure beyond raw tag lists. The reproducible pipeline supports portfolio demonstrations, research replication, and integration with image captioning outputs from the Flask application. Future work includes temporal drift analysis, multilingual tag normalization, and live API scoring for recommended hashtag bundles.

6. Reproducibility

Install dependencies with make setup, run make analyze to regenerate tables and figures, then make whitepaper to rebuild this document. Artifacts are written to outputs/tables, outputs/figures, and outputs/reports.